



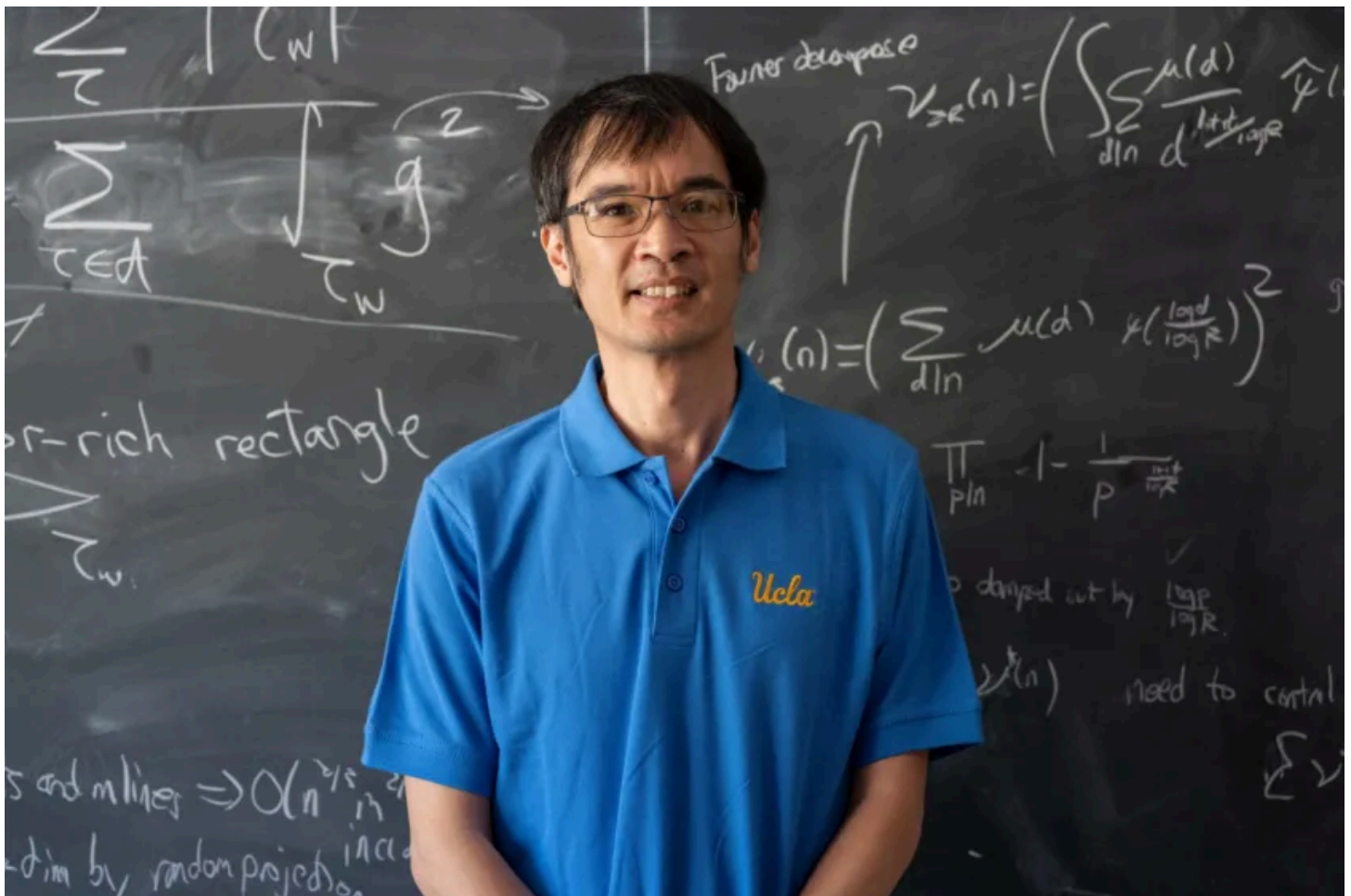
Mathematics

Terence Tao: AI companies are harming mathematics

AI firms are making new mathematical discoveries at a rapid pace, but not sticking around to help unpick the new proofs. That is not the way to advance our understanding, says mathematician Terence Tao

By **Matthew Sparkes**

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▲ Terence Tao is unimpressed with the way AI firms are approaching mathematics

David Esquivel/UCLA

Terence Tao is a man who understands mathematics. He has a Fields medal – often called the Nobel prize of mathematics – to name just one of his many accolades. So when

he says that OpenAI's recent blockbuster mathematical breakthrough is indicative of how big technology companies are negatively affecting the field, people are likely to listen.

This week, **OpenAI announced a solution to the Navier–Stokes problem**, one of the toughest and most enduring puzzles in mathematics. It is one of the Millennium Prize Problems, so solving it comes with a \$1 million reward. The news is the latest in a string of increasingly stunning and rapid advances in mathematics that have been building in recent months.

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But **Tao**, who is based at the University of California, Los Angeles, says the way in which discoveries are being made and “dumped” on the mathematical community may actually harm the field.

There has always been competition among mathematicians to be first, he says, which was fine because mathematics was so difficult that it created a natural brake to stop things getting out of hand. “Now, there's no speed limit, and suddenly things are breaking down,” he says.

Tao fears that new findings will keep landing at an increasing pace and that there will be no time to absorb the results, fully understand them, put them into textbooks and teach them to students. This means the staggering ability of AI to bring us new mathematics could actually harm the field, instead of boosting it.

Mathematics needs to overcome five hurdles in order to become useful, he says. First, it needs to be created, then checked, explained, accepted and finally digested into the corpus of work that is taught to the next generation. He says AI firms are only concerned with the first two.

“These companies are dumping carcasses of raw meat onto our communal village table and saying: ‘Here you go, I solved your food problem.’ And then they just leave,” says Tao.

As an example, he points to the situation surrounding **Tristan Buckmaster** at New York University and **Levent Alpöge** at AI company Anthropic. They had been “working slowly

on some very nice results” around Navier–Stokes, says Tao, and had developed a solution to the Euler equations, a close cousin of Navier–Stokes that would act as a stepping stone to solving it. But they say they were **forced to suddenly rush out papers** – which mathematicians told *New Scientist* lacked the clarity they would normally expect – once they found out that OpenAI had overtaken them.

Ironically, OpenAI only began working on the Navier–Stokes problem once it heard rumours of Buckmaster and Alpöge’s work.

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Tao says that what has developed is a race that benefits nobody (except perhaps the marketing departments of AI companies), erodes clarity and prioritises who is first over who is accurate, clear and concise.

“This is not how science is supposed to work,” says Tao. “They’re focusing on what traditionally has been the most prestigious part of mathematics, but then there’s sort of the lower-prestige work of explaining it, communicating it, teaching it, refereeing it for journals – all that work is left to us to clean up, and it’s demoralising.”

Increased competition from AI will also jeopardise the open, sharing culture that exists in mathematics and push research underground, he says, for fear of others swooping in on their ideas and completing them before they can publish.

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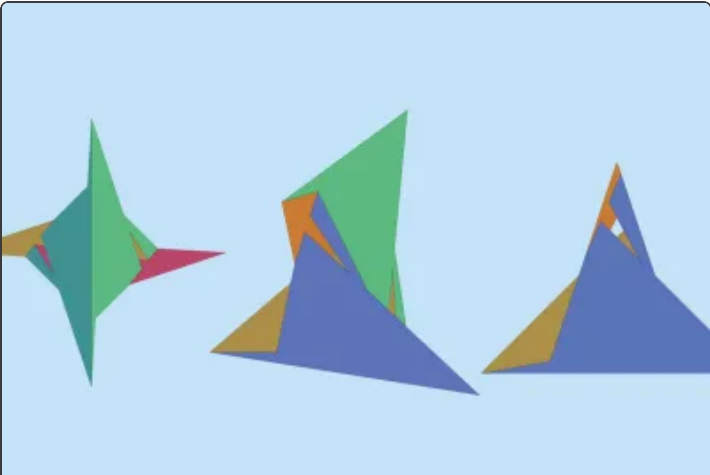
“It used to be that, if you announced you were working on a problem, there were maybe only a dozen people in the world who could plausibly outcompete you, and you knew all of them, you were on a first-name basis. There was no incentive to screw over your

colleagues because they would screw you back,” says Tao. “Knowing that someone’s working on a problem is now valuable information”

Despite his pessimism, Tao believes that mathematicians will eventually do the hard work of understanding, categorising and teaching AI company output. “It will eventually end up in textbooks, but it’s just a completely sordid way to get there,” he says.

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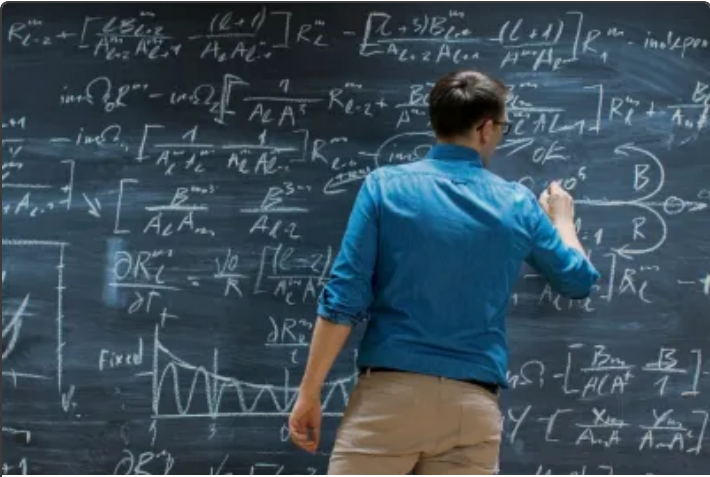
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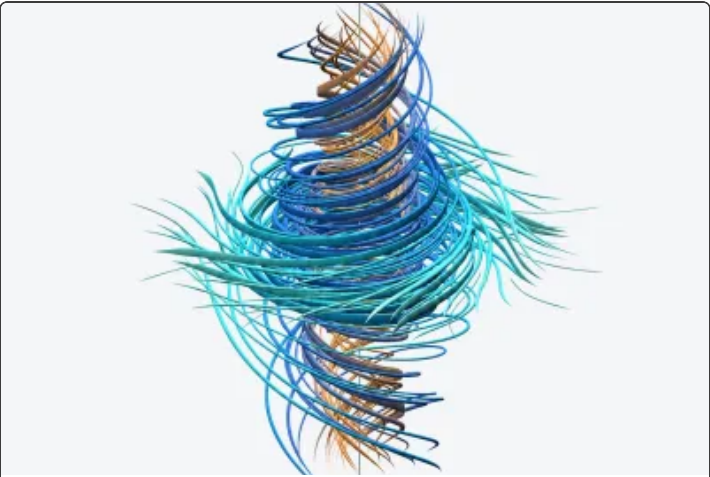
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